Programming Assignment III

Learning to Rank

Information Retrieval | Cranfield Collection

1. Group details

Group name: search_ninjas

Student ID	Member
23CS10002	Ananya Mondal
23CS10012	Charniya Krish
23CS30027	Ketan Suman
23CS30048	Shatakshi Singh
23EC37021	Shweta Agrawala

2. Problem statement

The objective is to learn a ranking function for the Cranfield collection using relevance judgments. Method A assigns weights to Boolean matches in four document fields. Method B uses a linear support vector machine over query-document retrieval features. The methods are compared with precision, recall, and F1 at fixed rank cutoffs.

The collection contains 1400 documents and 225 queries. The experiment uses the supplied Cranfield relevance judgments, rather than constructing new labels. Additional query files can be ranked without relevance judgments.

3. Implementation details

3.1 Data representation

Each document is split into title, author, bibliography, and body. The parser retains empty fields and concatenates repeated body sections. Text is converted to lowercase alphanumeric tokens and the scikit-learn English stopword list is removed; no stemming is applied. Document statistics are computed on the complete collection, which is known at retrieval time.

Cranfield query labels are non-contiguous, whereas the judgment file uses sequential query positions. The implementation records both identifiers. Judgment codes 1-4 are positive; code -1 and unjudged pairs are treated as nonrelevant. This is a binary relevance setting with incomplete judgments, not a claim that all unjudged pairs are truly irrelevant.

3.2 Method A: Boolean weighted zones

For a field f, its zone value is the number of distinct query terms present in f divided by the number of distinct query terms. The value is zero for an empty query. The document score is the sum of these four field values multiplied by their learned nonnegative weights. Thus each query term contributes a Boolean presence indicator even when it occurs repeatedly in a field.

Weights sum to one. An exhaustive search considers 286 combinations in increments of 0.1, selecting the highest mean training F1@10. For comparisons, zone_or uses a single any-term match indicator per field, and equal_zone uses equal field weights. Each model is trained independently where applicable.

Documents are ranked by the weighted field scores. When scores tie, the implementation uses increasing numeric document ID. The same ordering rule is applied in evaluation and prediction.

Field	Learned per-term weight	OR baseline weight
title	0.4	0.1
author	0.1	0.0
bibliography	0.2	0.1
body	0.3	0.8

3.3 Method B: linear SVM

The feature vector contains 22 values per document field, for 88 features in total. StandardScaler is fitted on training query-document pairs only. An SGDClassifier with hinge loss, L2 regularization, balanced class weights, and coefficient averaging fits a linear decision function. All training query-document pairs are used; there is no negative sampling. Documents are ranked by their signed decision values, without applying a classification threshold.

Feature family	Features computed for each of the four fields
Query coverage	Number of matched distinct terms and matched-term ratio
Field length and IDF	Token count and sum of IDF for matched terms
Raw term frequency	Sum, minimum, maximum, mean, and variance
Length-normalized TF	Sum, minimum, maximum, mean, and variance
TF-IDF	Sum, minimum, maximum, mean, and variance
Retrieval scores	Boolean OR match, vector-space cosine, and BM25

Out-of-vocabulary query terms contribute zero term frequency. Empty fields have zero matching, cosine, and BM25 features. The 88-feature definitions are unchanged between the original and revised zone experiments.

4. Plan of experiments

A seeded query-level split assigns 135 queries to training, 45 to validation, and 45 to test. All candidate documents for a query stay in the same partition. Method A weights are chosen on training F1@10. Method B's regularization parameter is chosen on validation F1@10 from 0.00001, 0.0001, 0.001, 0.01, 0.1, 1, 10, and 100. The evaluation SVM is retained without refitting.

For each query, precision is relevant documents retrieved divided by k, recall is relevant documents retrieved divided by all known relevant documents, and F1 is their harmonic mean. Each reported value is the arithmetic mean of per-query values. Cutoffs are 5, 10, 20, 50, and 100, with F1@10 used for selection.

The body-only BM25 and equal-zone rankings are untuned references. A separate final model is fitted to the 180 training and validation queries for scoring unknown queries. The final model does not contribute to the reported test metrics.

SVM alpha	Validation F1@10	Epochs
1e-05	0.1493	153
0.0001	0.1507	85
0.001	0.1713	74
0.01	0.1778	58
0.1	0.2160	21
1.0	0.2435	13
10.0	0.2381	46
100.0	0.2375	13

Selected alpha: 1. The evaluation scaler was fitted on 189,000 pairs, including 1,021 positive labels.

5. Results

5.1 Performance at rank 10

Method	Split	P@10	R@10	F1@10
zone	validation	0.2000	0.3266	0.2287
zone	test	0.1644	0.3329	0.2019
zone_or	validation	0.0267	0.0250	0.0242
zone_or	test	0.0356	0.0777	0.0447
svm	validation	0.2089	0.3592	0.2435
svm	test	0.1867	0.3600	0.2236
equal_zone	validation	0.1911	0.3163	0.2197
equal_zone	test	0.1733	0.3474	0.2109
body_bm25	validation	0.2111	0.3470	0.2430
body_bm25	test	0.1867	0.3522	0.2213

5.2 Test performance across cutoffs

Method	k	Precision	Recall	F1
zone	5	0.2178	0.2160	0.1945
zone	10	0.1644	0.3329	0.2019
zone	20	0.1211	0.4408	0.1773
zone	50	0.0613	0.5104	0.1042
zone	100	0.0380	0.6391	0.0696
zone_or	5	0.0400	0.0361	0.0334
zone_or	10	0.0356	0.0777	0.0447
zone_or	20	0.0433	0.1694	0.0644
zone_or	50	0.0333	0.2814	0.0557
zone_or	100	0.0269	0.4757	0.0492
svm	5	0.2489	0.2497	0.2258
svm	10	0.1867	0.3600	0.2236
svm	20	0.1244	0.4506	0.1818
svm	50	0.0627	0.5375	0.1070
svm	100	0.0407	0.6837	0.0744
equal_zone	5	0.2178	0.2352	0.2020
equal_zone	10	0.1733	0.3474	0.2109
equal_zone	20	0.1211	0.4496	0.1791
equal_zone	50	0.0627	0.5379	0.1067
equal_zone	100	0.0376	0.6359	0.0689
body_bm25	5	0.2533	0.2624	0.2335
body_bm25	10	0.1867	0.3522	0.2213
body_bm25	20	0.1333	0.4796	0.1925
body_bm25	50	0.0698	0.5932	0.1192
body_bm25	100	0.0400	0.6601	0.0732

5.3 Model diagnostics

Zone variant	Best weight tuples	Mean distinct scores/query	First test top-score ties
zone	1	19.8	1
zone_or	25	3.9	198

6. Discussion of results

The original OR-based zone method has low score resolution. At least one query term appears in 51.3% of document bodies, but its selected weighted score takes only 3.9 distinct values per query on average. For the first test query, 198 documents share the top score. Most of their relative order is therefore determined by the document-ID tie rule.

Matching individual query terms gives 19.8 distinct scores per query on average. On the test split, F1@10 increases from 0.0447 for OR zones to 0.2019 for per-term zones. This gain comes from representing partial query coverage, not from adding term frequency or BM25 to Method A.

Equal-weight zones achieve test F1@10 of 0.2109, compared with 0.2019 for the learned zone weights. The learned weights do not consistently improve this baseline on the test split; selection optimizes the training partition only.

On validation, the SVM scores 0.2435 F1@10 and body BM25 scores 0.2430. On test, their respective values are 0.2236 and 0.2213. The test difference of +0.0023 is small. The coefficients give weight to several lexical retrieval signals, but correlated features make coefficient magnitude an imperfect indication of individual feature importance.

The three largest absolute standardized SVM coefficients are body.vector_space (+0.0838), body.bm25 (+0.0813), body.idf (+0.0664).

There are important limits to interpretation. Only one split of a relatively small query collection is reported, and incomplete judgments are counted as negatives. The SVM optimizes pointwise hinge loss rather than ranked-list F1. Furthermore, the expanded regularization grid and per-term zone formulation were introduced after earlier test results had been inspected, so this is not a freshly untouched test evaluation. Larger independent query sets or repeated query-level cross-validation would give stronger evidence about generalization.

For unknown queries, the final model refits the SVM and zone weights on 180 training and validation queries. The 45 test queries remain excluded from fitting. The executable writes positional-ID and original-label TREC runs with the model scores and recorded ranks.

References

Programming Assignment III: Learning to Rank (course handout); Cranfield collection and relevance judgments (supplied files); scikit-learn documentation for StandardScaler and SGDClassifier; ir_datasets Cranfield relevance-code definitions.